



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

KEYNOTE SPEAKER REPORT SUBMISSION

Towards Trustworthy and Sustainable Agentic AI: Building Energy-Efficient AI Systems

Resource Person: Dr. Malka N. Halgamuge, Senior Lecturer in Cybersecurity, RMIT University, Melbourne, Australia

Submitted By: Sparsh Goswami (24070521091)



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About the Speaker

Dr. Malka N. Halgamuge is a Senior Lecturer in Cybersecurity at RMIT University, Melbourne, and a Senior Member of the IEEE. Her research covers cybersecurity, agentic AI security, IoT and the energy efficiency of large language models. She has over 120 peer-reviewed publications, has co-authored five books, and is listed among the top 2 percent of the world's most cited researchers (Stanford database, 2020 to 2024). She has given keynote and plenary talks at several international conferences and holds adjunct professorial roles in India.

Overview of the Session

The keynote looked at agentic AI, meaning systems that can take in information, make decisions, plan actions and work toward a goal with little human input. A conventional AI model usually performs one fixed task. An agent can plan, use tools, hold memory and carry out several steps on its own, adjusting as the situation changes.

Dr. Halgamuge argued that trustworthiness and energy efficiency both have to be designed in from the start, because neither can be added afterwards. She also connected two questions that are usually discussed separately: how to make autonomous AI safe, and how to make it affordable to run at scale.

Trustworthy AI and Human Oversight

The session named transparency, reliability, security, fairness, accountability and responsible use of data as the basics of trustworthy AI. Human involvement remains necessary, especially when an agent is used in sensitive or high-impact situations.

Sustainable and Energy-Efficient AI

Powerful AI models need a lot of computing power, and computing power needs energy. For agents the problem is larger, since every planning loop, tool call and memory read adds more computation. The approaches discussed were model optimization, efficient algorithms, removing unneeded calculations, choosing suitable hardware and using cloud resources well. A smaller, well-tuned model can often finish a task with fewer computational resources than a larger one.

Key Points

- **Agentic AI costs more to run and is riskier.** Every planning loop, tool call and memory read adds compute and a possible point of attack, so energy use and security exposure grow together.
- **Trust is more than accuracy.** A trustworthy agent needs safety, robustness, privacy, security and explainability. A good benchmark score alone does not show this.



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- **You cannot manage what you do not measure.** Energy and emissions reporting for AI research is still mostly voluntary and inconsistent, a gap the speaker's own 2026 study on mandatory disclosure points to.
- **Efficiency and safety can work together.** Right-sized models, fewer unnecessary reasoning steps, caching, and compression of data and context were suggested as ways to cut energy use without losing reliability.
- **Evaluation needs to catch up.** In the speaker's benchmark of two real agentic systems, OpenClaw and NanoBot, full completion was 26 percent for both, but NanoBot led on partial completion, 43 percent to 26 percent. System design matters as much as the underlying model.

My Takeaway

The session made it clear that building smarter AI agents is about more than capability. It also depends on whether they can be trusted and whether the planet can afford to run them. This connects to areas like data compression and networking, where sending and storing less data means using less energy. It also changed how I see efficiency: it is a design requirement from the start, not a cost saving added later.

Conclusion

The talk covered both the opportunities and the difficulties of autonomous AI. Making systems more capable is only part of the job. They also have to be trustworthy, safe, efficient and sustainable. For students who will go on to build these systems, that is a useful habit to start early.